

and they start looking for EMS partners to outsource it.

Unfortunately, this simple throw-it-over-the-wall approach neither helps the product owner nor the EMS partner. To understand the issues involved in electronic product manufacturing, let us try to see what happens inside an EMS partner's premises. Please see Fig. 1, which shows the following three discrete blocks of activities:

1. Supply chain
2. Manufacturing
3. Support and sustenance

Supply chain is the first element that impacts the product's success. It deals with the electronic components, custom parts (mechanical, plastic, electronics, etc), their quality, and their storage in the plant (we will see why storage is key a little later), all of which play a vital role.

Next is the actual manufacturing phase, which covers new product introduction (NPI), board and system assembly, testing of the PCBAs, and the final product (including calibration). Shipping is also important, as a product delivered in a damaged condition can result in customer dissatisfaction and expensive replacements or repair.

The final element is the product support. It covers both warranty and post-warranty repairs, obsolescence management, customisation of products for the global market, and global manufacturing.

These aspects show how important and intense the manufacturing is. It is not just the assembly of PCBs.

Why manufacturing is critical

Good manufacturing ensures a reliable long-lasting product. But for a successful product, it has to be designed such that the manufacturing line can easily adapt to the design. This is easier said than done. While startups and MSMEs focus on design and cost, they miss out on crucial manufacturing.

With increased use of surface mount technology (SMT) parts, automated assembly and manufacturing is inevitable. This essentially means the PCBs and the systems have to be assembled by contract manufacturers (EMS partners) who invest in automated lines and do manufacturing as

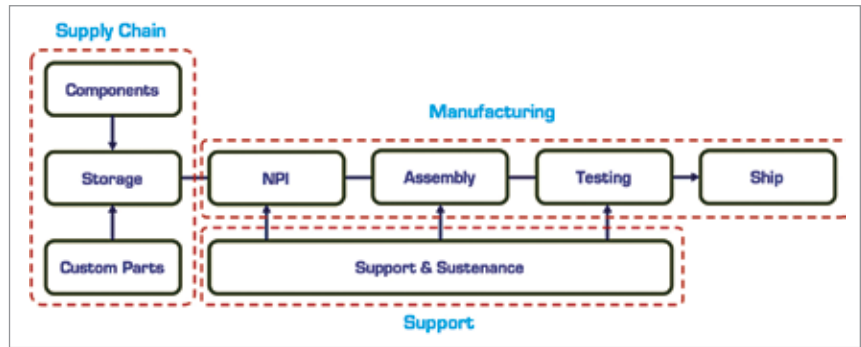


Fig. 1: Activities at an EMS

a service. This outsourcing imposes some ground rules for successful product manufacturing.

Let us have a look at the rules for better understanding:

1. The main process is the procurement of components and how these are stored. Sourcing components from an established distributor or a vendor is the most critical aspect, which is rarely followed. While buying components from local/unauthorised sources for the prototype may be fine, for bulk production these have to be procured from authorised sources.

2. Electronic components, in general, have an affinity for absorbing moisture. While for parts like transistors and diode-like items the impact of moisture is less, IC packages suffer the maximum damage. Generally, ICs that have absorbed moisture crack open during reflow soldering due to expansion of water vapour inside the package, leading to a damaged board. In addition, raw PCBs also absorb moisture, but EMS vendors bake the boards in an oven to ensure the PCBs are moisture-free. Absorption of moisture essentially forces the components to be stored in a dry environment and dehumidified storage space. Only established vendors have this facility as this type of storage is expensive.

3. Post storage, when the product gets into assembly line, there are three critical steps that decide the yield (good or bad) in the manufacturing line. However, the yield is decided by the design and is not a parameter that can be controlled by the EMS partner alone. For good PCB soldering, right quantity of solder, right component positioning, and right test access points are important. What complicates the issue is the dependence on

the type of manufacturing equipment used by an EMS partner.

To make the whole manufacturing process reliable and productive, almost all EMS partners have their own design guidelines and checklists to ensure high quality and good yield. These are called their DFX guidelines, where x stands for assembly, manufacturing, and testing.

The first guideline is a checklist known as DFM (design for manufacturing) process that defines the footprints of the components and how they are oriented. Typically, DFM has to be addressed early on during the PCB design stage itself. The DFM guideline ensures the right amount of solder is used for perfect reflow/wave soldering.

The second guideline is DFA (design for assembly), which ensures the components are located properly so that the pick and place machines place the parts 100% accurately. Also, aspects like multiple PCBs in one large PCB (known as paneling), tooling holes, and PCB silkscreens for legible reading are ensured.

Third and the most important is the DFT (design for test) guideline. It ensures proper testing of the assembled and soldered parts (PCBAs, subsystems, and final products). When the production volumes are high, EMS partners recommend in-circuit testing (ICT). This testing essentially accesses all the nodes in the PCB and tests the interconnections and the components.

Remember, ICT rarely tests the functionality of the system. This essentially means designers need to plan for the final product's testing. Also, in most cases, they need to design the functional testers (automatic test equipment known as ATE)